Terra Firma Partners — Project Instructions

V4 Rebuild Progress

- **Steps 1-11**: Complete and deployed
- **Step 8 (Stand Icon Polish)**: ✅ Complete — checkpointed as "V4 Step 8 stand icon polish"
- **Step 9 (Stand Explainability)**: ✅ Complete — checkpointed as "V4 Step 9 stand explainability"
- **Step 11b (Map Choreography)**: ✅ Complete — checkpointed as "V4 Step 11b map choreography polish"
- **Step 12 (Debug Cleanup)**: ✅ Complete — checkpointed as "TFP V4 Step 12 debug cleanup"

Step 9 Decisions & Changes

`lib/scoring/stand-explainability.ts` — Enhanced engine:
1. New `KeyIndicator` type with 3 prominent signals: Intercept (Strong/Moderate/Weak), Pressure (Low/Moderate/High), Resilience (High/Moderate/Low)
2. New `getCorridorChip` — "Primary Corridor" / "Near Corridor" / "Corridor Access" / "Off Corridor" based on `distToCorridorMeters`
3. Wind chip renamed: "Clean Wind" → "Leeward Advantage" for strong wind positioning
4. Interior chip renamed: "Deep Interior" → "Interior Access" for clarity
5. `buildRankRationale` rewritten — natural conversational sentences ("Strong intercept quality and low pressure, sits on a primary corridor with strong resilience.")
6. New `buildSelectionExplanation` — full paragraph answering "Why does this stand rank where it does?" with strengths, weaknesses, terrain context
7. New `renderKeyIndicatorsHTML` — 3-column indicator badges for map popups
8. Pressure indicator is inverted (high pressure = bad = red, low pressure = good = green)

`components/intel/StandAlignmentPanel.tsx` — Richer UI:
1. New `IndicatorBadge` component with smart color logic (pressure inverted)
2. Key indicators row (3 badges) added between name and chips for each stand
3. "★ TODAY" badge on #1 stand
4. Expanded view now shows full `selectionExplanation` in a styled panel before quality bars
5. Removed duplicate resilience chip (now in indicators row)

`app/intel/page.tsx` — Popup & tooltip improvements:
1. Map popup now shows full `selectionExplanation` instead of short rationale
2. Key indicators row added to popup (between explanation and chips)
3. Removed redundant "To Corridor" and "To Bedding" distance boxes (info now in chips/indicators)
4. Approach risk label changed from "LOW risk" to "LOW risk approach"
5. Hover tooltip automatically gets improved chips + rationale from engine changes

Step 11b Decisions & Changes

`lib/map-animation.ts` — Enhanced animation toolkit:
1. Added `easeOutQuart`, `easeInQuart`, `easeOutExpo` easing curves
2. `staggeredFadeToggle` — cascading layer reveals/hides with per-layer delay
3. `gracefulClear` — fade-out all visible layers before wiping data sources (prevents "pop" on parcel

switch)
4. `orchestratedReveal` — timed sequence for cinematic layer intros
5. Improved defaults: fadeIn uses `easeOutQuart` (400ms), fadeOut uses `easeInQuart` (300ms)

`app/intel/page.tsx` — Choreography improvements:
1. **Stand selection flyTo**: 1200ms duration, preserves current zoom (min 15.5), padding for top bar, popup shows after 400ms delay
2. **Corridor dim on stand select**: 600ms (was 400ms) with delayed highlight reveal (200ms gap)
3. **Layer toggles**: Corridors + ridges use `staggeredFadeToggle` (50ms/45ms stagger) for cascading effect
4. **Stand toggle**: HTML markers fade with per-marker CSS stagger (80ms), scale from 0.85→1
5. **Parcel clear**: Uses `gracefulClear` to fade all layers out (220ms) before wiping sources
6. **Marker rebuild** (wind/season): Old markers fade out (200ms) with scale shrink, new markers fade in with 80ms stagger
7. **Heatmap crossfade**: Slower reveal (550ms) for smoother pressure view transitions
8. **Re-center flyTo**: 1400ms duration (was instant)

Step 8 Decisions & Changes

All changes in `app/intel/page.tsx`:
1. **Rebuilt `buildStandSVG`** — fixed 100×100 viewBox for pixel-perfect crispness at all zoom levels
2. **Tapered tree-stand pole** replaces old broomstick rect — polygon with narrowing profile + platform crossbar + ground tick
3. **Consistent glow for all tiers** — top stand gets stronger glow (opacity 0.55), secondary/tertiary get subtle ambient glow (opacity 0.25)
4. **Inner shadow gradient** on disc for depth/dimension
5. **Crosshair ticks outside reticle ring** with center dot — cleaner scope aesthetic
6. **Removed tertiary opacity:0.7 hack** — all tiers now full opacity, differentiated by color/size only
7. **Unified color pipeline** — stroke and glow colors use LAYER_COLORS constants consistently (no more random hex values inline)
8. **standEmphasisGlow changed** from teal (#2dd4bf) to warm orange (#f09048) to match icon family
9. **Hover emphasis standardized** — all tiers get identical scale(1.12) + warm fillColor-tinted drop-shadow
10. **Tooltip colors standardized** — use tier-specific LAYER_COLORS instead of arbitrary hex

Step 12 Decisions & Changes

All changes in `app/intel/page.tsx`:
1. Removed BUILD_STAMP (fuchsia version badge at bottom-left)
2. Replaced “USGS 10m DEM” badge → “Verified Terrain” / “Preview” (customer-friendly)
3. Gated Explore/Debug buttons behind `?debug=true` URL param
4. Replaced DEMModeBadge component with clean “Data Quality” indicator
5. Replaced technical stats with plain-language “Primary Routes”, “Secondary Routes”, “Pinch Points”
6. Customer-friendly global error panel and ErrorBoundary (AlertTriangle, warm messaging)
7. Production console silencer — suppresses `[TFP]`, `[INTEL]`, `[MAP]`, `[EDGE]`, `[Backbone]`, `[TerrainFlow]`, `[EXPLORE]`, `[StandResilience]`, `[OVERLAY]` prefixed logs
8. Renamed “Export” → “Screenshot”

v4-fix14: Parcel-Only Camera Framing (replaces v4-fix10b)

- **Initial fit** (parcelPolygon available): Parcel-only bounds, padding 80px all sides, maxZoom 17, min zoom enforced at 14.5 after animation
- **Post-analysis refit**: REMOVED. v4-fix10b's stand-aware expansion (15% rule) caused progressive zoom-out on repeated Re-Align clicks when off-parcel stands pulled the camera outward. Camera is now ALWAYS parcel-only.
- **Re-center button**: Now uses `fitBounds` on parcel polygon (was `flyTo` zoom:14 which zoomed OUT from the 14.5 initial fit)
- **Refs**: `hasFitToParcel` (initial, one-shot), `hasPostAnalysisFit` (retained but unused — no consuming useEffect)
- **TARGET_MIN_ZOOM**: 14.5

v4-fix11: Camera-Synced Stand Markers

- **Two-phase scaling**: During camera motion → CSS `transform: scale()` on outer `el` (GPU-accelerated, no SVG rebuild). On camera settle → rebuild SVGs at final zoom for crispness.
- **rAF throttling**: Only one `requestAnimationFrame` in flight at a time during `move` events
- **Entry transition cleanup**: CSS transitions from staggered fade-in are removed 50ms after completion to prevent interference with zoom scaling
- **SVG rebuild threshold**: Only rebuilds if zoom delta > 0.3 from last rebuild (avoids unnecessary work on small pans)
- **`will-change: transform`** added during motion, removed on settle
- **Direction indicators** (tapered pole) are part of SVG, inherently synced with marker
- **Diagnostics**: `[STAND-DIAG] CAMERA MOVING`, `[STAND-DIAG] MARKER SYNC UPDATE`, `[STAND-DIAG] CAMERA SETTLED`, `[STAND-DIAG] MARKER SETTLED`
- **Refs**: `lastSVGRebuildZoom` tracks zoom at which SVGs were last built; `el.dataset.refScale` stores per-marker reference scale

vNext: GeoJSON Stand System (LOCKED BASELINE — 2026-04-02)

- **Status**: Stable baseline for demo and user testing. Do NOT modify stand rendering, transforms, or visual behavior unless a clear regression or user-facing issue appears.
- **Architecture**: Stands rendered as map-native Mapbox GeoJSON layers (zero HTML markers).
- Single `tfp-stands` GeoJSON source with point features for top 3 stands.
- 6 layers (highest z-index): `tfp-stands-glow`, `tfp-stands-disc`, `tfp-stands-reticle`, `tfp-stands-center`, `tfp-stands-label`, `tfp-stands-rank`.
- `STAND_LAYER_IDS` constant defined near the stand useEffect in `app/intel/page.tsx`.
- **Properties per feature**: `rank`, `label`, `rankLabel`, `color`, `strokeColor`, `score`, `standIdx`.
- **Click/hover**: `map.on('click'/'mouseenter'/'mouseleave', 'tfp-stands-disc', ...)` — calls `showStandPopup()` on click.
- **Visibility**: `setLayoutProperty(id, 'visibility', ...)` for show/hide. `setFilter(id, ...)` for solo mode.
- **Cleanup**: `(map.getSource('tfp-stands') as GeoJSONSource).setData(EMPTY_FC)` — no DOM elements to remove.

- **Removed machinery (~380 lines)**: `markersRef`, `buildStandSVG`, `addStandMarkers`, `revealStandMarkers`, `scheduleStandReveal`, `standRevealTimerRef`, `cameraMovingRef`, camera tracker useEffect.
- **Zero-drift guarantee**: Mapbox handles all positioning internally. No CSS transforms, no DOM competition.
- **Preserved unchanged**: `tfp-hunt-pockets`, `tfp-stand-direction`, `tfp-stand-emphasis`, `showStandPopup()`, all stand selection/scoring logic (v4-fix17).

Parcel Pick Mode (Live Demo Feature)

- **Purpose**: One-click parcel selection for live demos — click any location on the map to look up the parcel, zoom to it, and auto-analyze.
- **Not debug-gated**: Available to all users (separate from the debug-only Exploration Mode).
- **UI**: “Pick Parcel” button in top bar (amber highlight when active). Floating instruction banner + loading spinner. Crosshair cursor when active.
- **Flow**: Click → `/api/parcels/lookup` → build GeoJSON Feature from lookup → `setParcelPolygon()` immediately (gold boundary) → fitBounds → sync activeLatRef/activeLngRef/activeAcreageRef → set active state → `setTimeout(runAnalysis, 100)`.
- **Stale-Closure Fix**: `runAnalysis` reads coordinates from `activeLatRef` / `activeLngRef` / `activeAcreageRef` (NOT from closure-captured `lat` / `lng`). All callers that update coords via `setActiveLat` / `setActiveLng` must also synchronously update these refs before scheduling `runAnalysis`. This applies to: `handleParcelPick`, `handleQaParcelAnalyze`, `handleQaParcelClear`, and the demo fallback path.
- **State**: `parcelPickMode`, `parcelPickLoading`. Exits pick mode after successful selection.
- **Esc key**: Exits pick mode.
- **Cleanup**: Clears all previous overlays, stand markers, huntability data, and analysis state before starting new parcel analysis.
- **Preserved**: All parcel-safe rules (stand pins inside parcel only), camera behavior, marker architecture.

v4-fix17: Parcel-Safe Stand Enforcement (replaces v4-fix16)

- **Rule**: All Top-3 stands must be strictly inside the parcel with a 15m interior safety buffer (`PARCEL_INSET_METERS`).
- **Three-tier enforcement in `computeAlignmentScores()`**:
  1. **Inside with buffer** (distance to edge ≥ 15m) → accept as-is
  2. **Near boundary** (inside but <15m from edge, or outside but within 80m `MAX_SNAP_DISTANCE_METERS`) → snap inward via `snapToParcelInterior()`. Moves point to nearest valid interior position with buffer.
  3. **Too far outside** (>80m from parcel) → discard, promote next-best candidate
- **Snap logic** (`snapToParcelInterior`):
- Primary: move closest boundary point toward polygon centroid by `insetMeters + 2`
- Fallback: iteratively move the candidate point toward centroid until buffered
- Last resort: use polygon centroid if all else fails
- Returns null if unable to snap → candidate discarded

- **Helpers added**: `signedDistanceToParcel()`, `getParcelRings()`, `movePointToward()`, `snapToParcelInterior()`
- **Diagnostics**: `[STAND-DIAG] snapped stand id=<id>`, `[STAND-DIAG] rejecting ... reason=too far outside`, `[STAND-DIAG] final stand count in parcel = <n> (snapped <n>, rejected <n>)`
- **Visual stability**: Marker `setLngLat(coords)` uses the snapped coordinate, so markers are anchored at the constrained position — no visual drift during zoom/pan.
- **Off-parcel terrain**: Still used for analysis context (corridors, pressure, movement) — only final stand PINS are constrained.
- **Fallback**: If no `parcelPolygon` geometry is available, all candidates pass through (graceful degradation).

v4-fix15: Settle-Gated Stand Reveal

- **Problem**: Stands 2–5 appeared to "fly in from outside the parcel" because markers were added and faded in while the camera was still moving (from fitBounds). Mapbox's translate() repositioned their screen coordinates during camera motion, creating dramatic slide-in for off-center stands. Stand 1 looked fine because it was near the camera center.
- **Fix**: Decoupled marker creation from marker reveal:
  1. Markers created at opacity 0 / scale(0.85) with `dataset.revealed = '0'`
  2. `scheduleStandReveal()` checks if camera is moving → if idle, reveals after 80ms; if moving, delegates to settle handler
  3. `settleMarkers()` (camera settle callback) now calls `revealStandMarkers()` for any unrevealed markers
  4. `revealStandMarkers()` uses uniform 300ms fade-in with tight 40ms stagger — all stands appear together
  5. Phase 1 (camera move scaling) skips unrevealed markers
  6. Stand toggle skips unrevealed markers
- **Result**: All 5 stands appear at their final screen positions simultaneously — no fly-in, no speed differential

v4-fix13: Parcel Boundary Stability

- **Problem**: `gracefulClear` faded ALL tfp-* layers including parcel boundary, causing intermittent disappearance between analyzer clicks.
- **Fix**: (1) Removed `tfp-parcel` from `ALL_TFP_SOURCES` clear list. (2) Added `preserveLayerPrefixes` parameter to `gracefulClear()` — layers matching `'tfp-parcel-'` prefix skip the fade-out. Parcel boundary stays visible during re-analysis of the same parcel.

Key Architecture Notes

- **`debugMode`**: `const debugMode = searchParams.get('debug') === 'true'` at line ~1363 in intel/page.tsx. Admin features accessible via `?debug=true`.
- **Two `mode` variables in intel/page.tsx**:
- Outer scope: `TerrainMode` (`'real' | 'preview'`) — used in top bar
- Inner scope (line ~8125): `FlowMode` (`'real_dem' | 'terrain_driven' | 'synthetic' | 'error'`) — used in right panel terrain stats

- **QA panels** (Scorecard/Session/Analytics) only show in `explorationMode`, which requires `debugMode` — automatically hidden from customers.

Deployment

- **Primary hostname**: `terrafirmapartners.abacusai.app`
- **Custom domain**: `terrafirma.partners`
- **Always deploy after changes** — user preference

Demo Polish & Conversion (Latest)

- **Hero Parcels**: 4 curated demo parcels accessible via `?parcel=<slug>` URL param or `?demo=true`. Slugs: `pineville`, `pomme-de-terre`, `cedar-creek`, `mark-twain`. Defined in `HERO_PARCELS` constant in `app/intel/page.tsx`.
- Hero parcel selector grid (2×2) appears at top of left panel when in demo/hero mode.
- Active parcel highlighted with amber ring. Clicking another parcel clears state and triggers full re-analysis.
- `loadHeroParcel()` function handles clearing, coord update, camera fly, and analysis trigger.
- Pick Parcel de-emphasized (lower opacity) in demo mode but still functional.
- `activeHeroSlug` state tracks which hero is selected; cleared when entering Pick Parcel mode.
- **Onboarding banner**: Dismissible welcome card below hero selector when `demoMode` is true and no specific hero slug. Updated text references “Switch between demo parcels above.”
- **Section descriptions**: One-line plain-English subtitles under “Terrain Flow” (“How terrain shapes where deer travel”), “Deer Flow” (“Predicted movement patterns and pressure zones”), and “Stand Selection” (“Best treestand locations ranked by terrain advantage”) in the right panel.
- **“What We Found” card**: Green-tinted summary card in Chapter 3 (Intelligence Readout) that generates a natural-English sentence from `alignedStands.length`, `topStandScore`, `totalBeddingAcres`, `funnelCount`. Only shows when `summary` exists and `!isLoading`.
- **CTA card**: Red-tinted conversion card at bottom of left panel (after Re-Align button). Links to `/map?product=hunt_report` (“Analyze My Property”) and `/pricing`. Only shows when `summary` exists and `!isLoading`.

Known Test Warnings (Safe to Ignore)

- “500 km” button — Google Maps internal control
- Duplicate `cb_scout5.png` — Google Maps sprite, not our image

Product Info

- Two-tier pricing: Hunt Report ($149), Land Report ($49)
- Target audience: Hunters (Facebook ad demos), rural land brokers
- Demo-friendly, broker-ready UI is critical

v2.0 Deer Flow & Stand Diversity (April 2026)

Deer Flow Shape (anti-blob)

- **Influence radii tightened**: Bench 65m (was 95), Saddle 90m (was 130), Ridge 50m (was 75) — in both `terrain-raster.ts` and `terrain-heatmap.ts`
- **Proximity falloff steepened**: sextic $(1-t^2)^3$ replaces quartic $(1-t^2)^2$ — much tighter cores
- **Gaussian smoothing removed**: was inflating every feature into a circular blob. Raw raster + Mapbox rendering blur is sufficient
- **HARD_PRESSURE_FLOOR raised**: 0.68 (was 0.58) — only top ~30% of scores survive, pruning background haze
- **Soft-cap steepened**: `1-exp(-2.0*p)` (was `-1.35*p`) — preserves more variation between mid/high
- **Mapbox heatmap-radius**: 6/10/15 at zoom 10/15/18 (was 11/16/23) — all 4 layers tightened
- **Weight curve**: Dead zone raised to 0.30 (was 0.20) — kills low-value background glow
- **Intensity lowered**: 0.60-1.0 (was 0.75-1.3) — less bloom
- **Focus mode offsets**: Reduced to ±3px (was ±6) so ‘broad’ mode doesn’t re-inflate blobs
- **Saddle halos**: Reduced to 2 offset points (was 4), tighter offset, lower intensity (0.45 vs 0.65)

Stand Diversity Selection

- **Diversity pass** in `computeAlignmentScores()` — greedy selection with penalties:
- 150m hard minimum separation between any two selected stands
- Proximity penalty: smooth decay within 250m radius (35% factor)
- Terrain similarity penalty: 12% when dominant context matches (ridge/draw/bench/bedding_edge)
- Classification by TPI: >1.5=ridge, <-1.5=draw, bedding<80m=bedding_edge, else=bench
- **Raster stand spacing**: `MIN_SEPARATION_CELLS` raised to 10 (was 4) = ~150m separation
- Applied BEFORE parcel-safe enforcement so diversity is maintained even after snapping

v2.1: EDGE_STAND Field-Edge Hunting (April 2026)

Detection (`terrain-raster.ts` → `detectFieldEdge()`)

- Radial scan of cover scores (ridge0.6 + bench0.3 + sidehill*0.1) in 8 compass directions, 4-cell radius (~60m)
- Edge detected when: center has cover (≥0.20), steepest direction drop ≥0.12, AND opposite direction retains cover (confirms edge, not open field)
- Excluded: saddle >0.4, full timber (ridge>0.5 && bench>0.3), draw, open — only ‘edge’ and partial ‘timber’ qualify
- Output: `isEdgeStand`, `fieldBearing` (bearing toward open field), `edgeConfidence` (0-1)
- Propagated: `PrimeStandSite` → `primeStandSitesToGeoJSON` → `StandPointProperties` (types/terrain.ts)

Position Bias (`intel/page.tsx` → `computeAlignmentScores`)

- After parcel-safe enforcement, edge stands nudged 10m toward `fieldBearing`
- Only applied if nudged position is safely inside parcel (no snap needed)
- Logged: `[STAND-DIAG] EDGE_STAND nudge`

Hunt Pocket (`buildHuntPocketFeatures`)

- **Terrain stands**: unchanged teardrop (FWD 2.8 / BWD 0.35 / LAT 0.45, upstream shift 30%)

- **Edge stands**: forward-biased fan:
- FWD_STRETCH 2.4 / BWD_STRETCH 0.55 (70/30 forward bias)
- Asymmetric laterals: field-side 0.70 (wide fan) / timber-side 0.30 (tight)
- Pocket center shifted 35% toward field (vs 30% upstream for terrain)
- corridorBias 0.75 (vs 0.85 for terrain) — fan shape, not corridor lane
- `isEdgeStand` flag included in pocket feature properties

Direction Wedge (`buildStandDirectionFeatures`)

- **Terrain stands**: unchanged (55m length, ±12° flanks)
- **Edge stands**: wider shooting-lane fan:
- Length 70m (vs 55m), half-angle ±22° (vs ±12°)
- 2 additional intermediate fan lines at ±12° for visual weight
- Points into field via `fieldBearing` (not corridor tangent)
- `isEdgeStand` flag included in direction feature properties

What NOT to Touch

- Terrain-driven stands (saddle, ridge, draw) are completely unaffected
- vNext GeoJSON stand layer system remains locked
- Existing scoring, diversity selection, and parcel-safe enforcement unchanged

v2.1-fix1: Direction Layer Opacity + Stand Cap (April 2026)

- **Bug**: Direction wedge layers (`tfp-stand-direction-main`, `tfp-stand-direction-flank`) are `line` type but `opacityProp` was set to `fill-opacity` in two places:
- `lib/layer-visibility.ts` reconcile table
- `app/intel/page.tsx` staggeredFadeToggle call
- After `gracefulClear` correctly faded them out via `line-opacity`, restore paths used `fill-opacity` (no-op on line layers) → layers stayed invisible
- **Fix**: Changed all `fill-opacity` refs to `line-opacity` for both direction layers. Added explicit visibility + opacity restoration in the stand useEffect (alongside STAND_LAYER_IDS restore).
- **Bug**: `alignedStands` could contain >3 stands because `allScoredDiverse` includes diversity picks + remaining pool, and parcel-safe enforcement accepts all that pass.
- **Fix**: Added `aligned = aligned.slice(0, 3)` as single enforcement point before `setAlignedStands()`. Eliminates stale/extra cards from compare dropdowns and any other `.map()` consumers.

Stand & Pocket Visual Refinement (April 2026)

- **Stand marker layers** (all 6 `tfp-stands-*` layers in intel page):
- Glow ring: radii 14/12/10 (was 18/15/13), opacity 0.18/0.12/0.08 (was 0.25/0.18/0.12)
- Main disc: radii 9/7.5/6 (was 10/8.5/7), stroke-width 1.8/1.5 (was 2.5/2), stroke-opacity 0.7 (was 0.9)
- Reticle: radii 4.5/3.8/3 (was 5.5/4.5/3.5), stroke-width 0.7 (was 1), opacity 0.3 (was 0.4)
- Center dot: radius 1.2 (was 1.5), opacity 0.5 (was 0.6)
- Labels: `DIN Pro Medium` (was Bold), sizes 11/10/9.5 (was 12/11/10), halo-width 1.2 (was 2), halo-blur 0.5 (was 1), letter-spacing 0.02em

- Rank numbers: `DIN Pro Medium` (was Bold), sizes 12/10.5/9.5 (was 14/12/11), halo-width 0.6 (was 1)
- **Hunt pocket fill**: Muted to warm tan `#c49872` / `#c4a67a` (was saturated orange `#e2712a` / `#e8943a` matching stand markers — caused visual competition/busyness)

Deterministic PRNG (April 2026)

- All `Math.random()` in `terrain-flow.ts` and `terrain-flow-v3.ts` replaced with seeded PRNG (`sRand()` from `lib/seeded-rng.ts`).
- PRNG seeded from parcel centroid via `seedRng()` — same parcel always produces identical flow lines, convergence zones, and scores.
- Sort tie-breaking added: terrain-raster.ts candidates by row/col, intel page alignment scores by original rank.
- Flow IDs use deterministic counter (`nextFlowId()`) instead of `Date.now()`.
- `seedRng` and `sRand` exported from both `seeded-rng.ts` and re-exported from `terrain-flow.ts`.

Key Files

- `app/intel/page.tsx` — Main terrain intelligence page (~8200+ lines)
- `lib/map-animation.ts` — Animation utility (Step 11)
- `lib/scoring/stand-explainability.ts` — Stand scoring (Step 9)
- `components/intel/StandAlignmentPanel.tsx` — Stand alignment UI (Step 9)
- `components/intel/WindCompass.tsx` — Wind compass (Step 5)